

Project Name	Allixir Intelligence
Online team meeting	https://tu-berlin.zoom-x.de/j/68161739941?pwd=Tz3duoAQeQHGWAsBNMexRqNxlnSPCl.1
Production system (if any)	...
Test system (if any)	...
GitHub repository	https://github.com/amosproj/amos2026ss03-ailixir-intelligence
GitHub feature board	https://github.com/orgs/amosproj/projects/99
GitHub imp-squared backlog	https://github.com/orgs/amosproj/projects/103/views/1
Position of T-shirt logo	Upper Torso Right Small
Link to logo for black T-shirt	https://www.shirtinator.com/s/w1QJzfDfTC-gGBjVmnCscg
Link to logo for white T-shirt	https://www.shirtinator.com/s/sMNil0MoQzmuNuc04Y6Ljw
Additional materials	...
Team mailing list	oss-amos-proj3@lists.fau.de

[illegible]

		Product Owner					
#	Meeting Day	Review	Planning	Software Developer	Release Manager	Scrum Master	Comment
1	2026-04-15		Eloi	Everyone else	N/A	COACH student	
2	2026-04-22	Eloi	Tobias	Everyone else	Andrew Ebied	COACH student	
3	2026-04-29	Tobias	Eloi	Everyone else	Andrew Ebied	COACH student	
4	2026-05-06	Eloi	Tobias	Everyone else	Hasnat Ahmed	COACH student	
5	2026-05-13	Tobias	Eloi	Everyone else	Hossam Amer	COACH student	
6	2026-05-20	Eloi	Tobias	...	Muhamad Zeeshan	COACH student	
7	2026-05-27	Tobias	Eloi		Hannes Vinzens	COACH student	
8	2026-06-03	Eloi	Tobias		Marcel Vogt	COACH student	
9	2026-06-10	Tobias	Eloi		Abdul Haseeb	COACH student	
10	2026-06-17	Eloi	Tobias		Hannes Vinzens	COACH student	
11	2026-06-24	Tobias	Eloi		Hasnat Ahmed	COACH student	
12	2026-07-01	Eloi	Tobias		Hossam Amer	COACH student	
13	2026-07-08	Tobias	Eloi		Muhamad Zeeshan	COACH student	
14	2026-07-15	Eloi	/		Abdul Haseeb	COACH student	Demo day!
15	2026-07-22		/			COACH student	Retrospective
Product owners, software developers, and Scrum Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here							

Goals	Deliver MVP
	mutual and personal improvement
	deliver a maintainable prototype that can be potentially used as a starting point for the industry partner
Meeting norms	Meeting Should stick to deliverables.
	maybe once or twice per week. for each meeting, a document should be made that includes the points which we will go over. the document could be edited by anyone in case anyone has points they want to discuss in the meeting, although this might be particularly useful for POs.
	write an email in case you will be late/unable to attend team meetings
	be on time
	mute yourself in calls if you don't talk
	Protocol industry partner meetings
	Team Meeting starts at 12 (30 minutes earlier)
Working norms	
	deciding on a release manager should be part of the team meetings
	Would be cool if we could give an estimate of when we will be available to work, and if possible, also specify on which issues and till when we expect to be finished with them. in case of delays, i think they should be communicated as well so that the team gets a rough idea of when they could be done or if help is needed from other people
	SDs can self-assign issues at will; if that doesn't work out, issues will be assigned during team meetings
	Coding (style) guidelines, common AI agent setup, CI/CD linting/formatting checks
	Documentation (how to run + code comments)
	Testing the written code/feature. All tests should pass before merging to main/master
	git flow strategy for working on code
	Definition of Done (DoD) A checklist of criteria (e.g. tests passed, code reviewed, documentation updated) that must be met before an issue is considered finished
	Definition of Ready (DoR); whether or not devs have enough capacity to take on an issue and finish it in time
	Pull Requests need to be reviewed and approved by at least one other developer
Coordination norms	
	Bigger decisions will be put to vote (ideally on slack)
	Work should be distributed with clear boundaries.
	use feature board actively and communicate with other people to coordinate work
Communication norms	
	replies within ~24h
	would also be cool if people communicate if they are unable to reply within 24h, with an estimate of when they will reply
	Communication with the industry partner over Slack

	it would also be nice to have dedicated communication channels for the team internally and with the industry partner. another dedicated channel for SDs as well
	agree on one main channel of communication: Slack
	Respect, active listening, tolerance
Consideration norms	
	Respect, Communicate Conflicts
	be kind and be cooperative
Cont. improvement norms	constructive criticism
	open feedback about things that could be improved
Rewards	
	eat ice cream together at the end of the course
Sanctions	
	gets brought up in the meeting
Signatures	
Scrum Master	Jonas Brüggemann
Product owner	Eloi Sandt
Product owner	Tobias Busse
Software developer	Hasnat Ahmed
Software developer	Hossam Amer
Software developer	Hannes Vinzens
Software developer	Muhammad Zeeshan
Software developer	Abdul Haseeb
Software developer	Andrew Ebied
Software developer	Marcel Vogt

Product Vision	Project Mission
Allixir Intelligence transforms unstructured documents into connected, actionable knowledge. Users can upload or scan documents (e.g., medical reports, invoices), from which the system extracts domain-specific, meaningful data fields and organises them into a personalised knowledge base. An AI-powered voice/chat interface enables users to ask precise, context-aware questions and better understand their personal documents, as well as relationships and developments across their document history. At its core, the platform is designed to be domain-agnostic and configurable for a wide range of use cases.	The mission of this project is to create an MVP that runs on mobile devices. Core functionality includes extracting structured information from uploaded documents, storing and organising it within a personal knowledge base, and enabling conversational querying using RAG with optional enrichment from external knowledge sources such as research papers. The system will demonstrate domain configurability using medical documents as the primary domain and finance as a secondary domain.

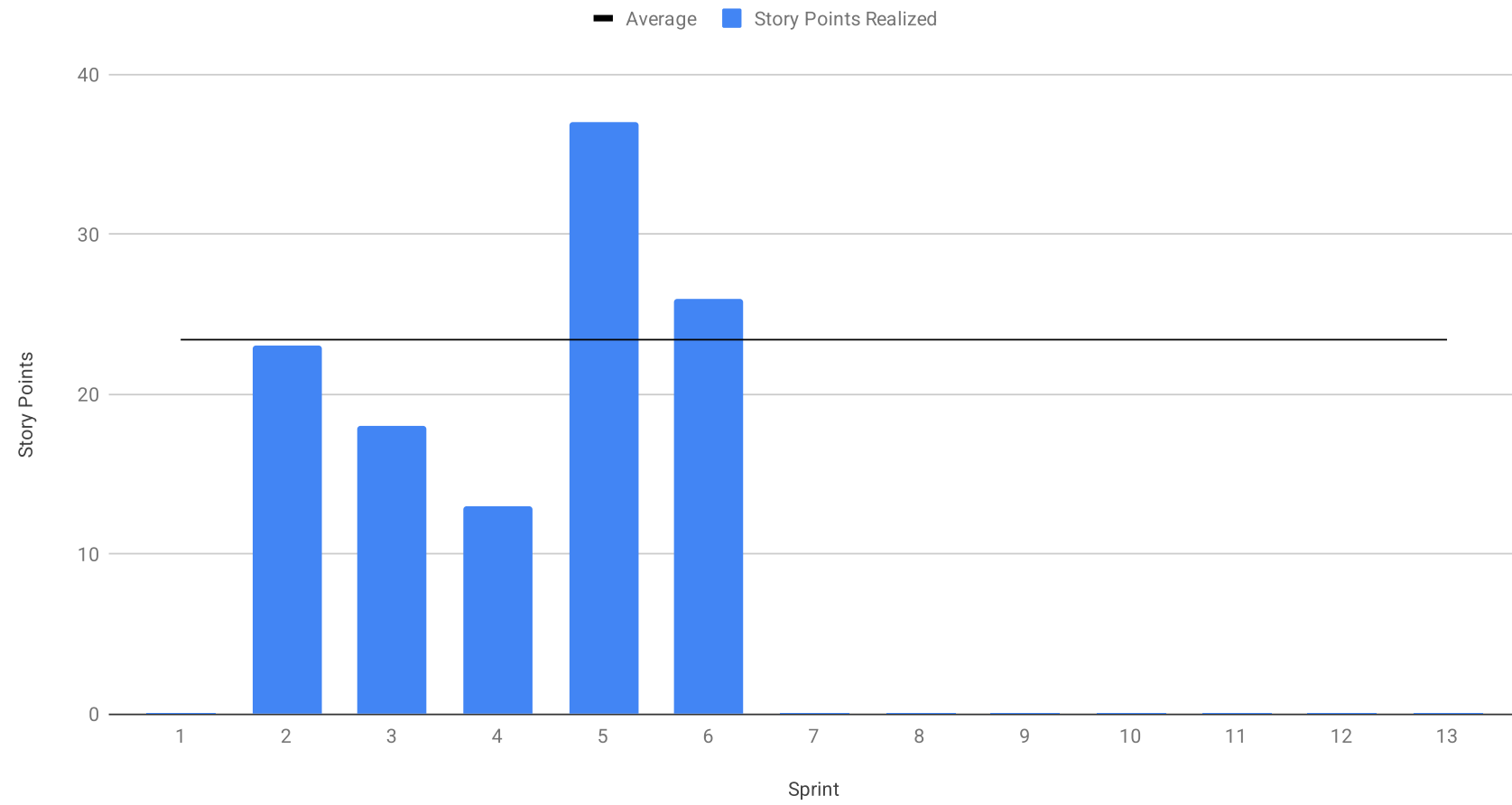
Term	Definition
AiLixir	A mobile application that lets users capture or upload documents, extract structured knowledge from them, and ask an AI questions about the analysed content.
User	A person who uses AiLixir to manage documents, start knowledge extraction, view results, and interact with the AI.
Account	A user identity that stores private app data and enables access to personalised documents, chats, and extracted knowledge.
Registration / Sign-up	The process by which a new user creates an account, typically using email, password, and required personal details.
Login	The process by which an existing user proves their identity and gains access to private app screens and backend functionality.
Authenticated User	A user who has successfully logged in and is allowed to access private documents, protected screens, and authenticated backend APIs.
Document	A user-provided file or scan that can be stored, previewed, analysed, and used as the source for knowledge extraction.
Captured Document	A document created by taking a picture or scan with the smartphone camera and passing the result back to the app.
Uploaded Document	A document selected from the device file system or submitted from the camera that becomes available inside AiLixir.
Document Preview	A small visual representation of a selected document or document page before the user starts analysis or extraction.
Document View	The screen that lists the documents stored for the authenticated user and acts as the entry point to document actions.
Document Details View	The screen for one document that displays metadata, pages, status, and actions such as starting extraction or downloading results.
Document Metadata	Descriptive information about a document, such as status, size, filename, document date, upload date, facility, and tags.
Document Page	An individual page or image belonging to a document, especially relevant for multi-page uploads or camera scans.
Document Tag / Label	A user-facing classification attached to a document, for example a lab result label, used to organize and filter documents.
Document Status	The current processing state of a document, such as not extracted, extraction in progress, or analysed.
Knowledge Extraction	The user-triggered process that converts document content into structured domain information, such as entities and relationships.
Extraction Status	An indicator showing whether knowledge has already been extracted from a document, is still being extracted, or has not been extracted yet.
Medical Document	A document containing medical information, such as a lab report, discharge summary, diagnosis information, medication information, or blood marker data.
Blood Biomarker	A measurable health value extracted from medical documents and displayed over time in the medical dashboard.
Medical Dashboard	A visual overview that shows extracted medical data, including biomarkers, historical trends, and filterable health information.
Financial Metric	A financial value extracted from documents or connected sources, such as balances, asset values, transaction data, or portfolio performance data.
Finance Dashboard	A visual overview that consolidates financial metrics and transactions and presents portfolio performance over time.
Structured Data	A machine-readable representation of extracted information, typically JSON or graph data, that downstream services can process reliably.
Domain-specific Entity	A meaningful object identified in a document, such as Patient, Diagnosis, Medication, MedicalReport, or a financial transaction.
Relationship	A typed connection between two domain-specific entities, such as HAS_DIAGNOSIS, PRESCRIBED_FOR, MENTIONS_DIAGNOSIS, or TREATED_BY.
Triple	A structured fact consisting of a subject entity, a relationship, and an object entity, prepared for knowledge graph ingestion.
Knowledge Graph	A structured network of entities and relationships extracted from documents and used to support retrieval, validation, visualisation, and LLM answers.
Node	An entity stored in the knowledge graph, such as a patient, diagnosis, medication, or document.
Edge	A relationship stored in the knowledge graph that connects two nodes and expresses how they are related.
Graph Database	Persistent storage for knowledge graph nodes and edges, implemented either as a dedicated graph database or another schema that can store triples.
Graph Ingestion	The act of writing extracted nodes and edges into the knowledge graph database so that they can be retrieved later.
Subgraph	A relevant subset of the knowledge graph returned for a specific query or prompt, usually consisting of targeted nodes and their direct edges.
Graph Retrieval	The process of finding and returning a query-relevant subgraph so that another service, especially the LLM integration, can use it as context.
LLM	A language model that generates answers for users and can use retrieved graph data as grounding context.
AI Agent	The conversational assistant in the app that receives user prompts and returns answers about uploaded and analysed documents.

Term	Definition
Prompt	A user input to the AI agent, typed or spoken, that asks for information or analysis about uploaded and analysed documents.
LLM Context	Structured information provided to the LLM before answer generation so that answers can reference verified extracted facts.
JSON Context Window	A JSON-formatted representation of graph facts that is injected into the LLM prompt for initial integration and validation.
Hallucination	An AI answer that is not supported by verified extracted facts; the knowledge graph is intended to reduce this risk.
GraphRAG Framework	A framework-based approach for extracting, storing, and retrieving graph-based knowledge for use with generative AI.
Manual JSON Pipeline	An extraction approach in which entities and relationships are first produced as structured JSON before being sent to an ingestion service.
Embedding	A vector representation of data sources such as PDFs, images, JSON files, or knowledge graphs, used for retrieval in a RAG setup.
RAG Database	A retrieval database that stores embeddings and supports finding relevant information for AI-generated answers.
Knowledge Graph Visualisation	A visual representation of a document's extracted graph, typically as a node-link diagram or structured report.
Knowledge Graph PDF Export	A generated PDF representation of a document's knowledge graph that the user can download and inspect locally.
Chat Screen	The app screen that displays user prompts and AI-agent responses and provides input controls for text and voice interaction.
Voice Dictation	A microphone-based input mode that turns spoken language into a text prompt for the AI agent.
Voice Conversation	A voice-based interaction mode in which the user can converse with the AI agent without relying only on typed prompts.
Backend API	A service interface that allows the frontend to authenticate users, store and retrieve documents, trigger extraction, and request generated results.
Document API	Backend functionality that accepts document uploads, returns the user's stored documents, and retrieves a document by document ID.
Persistent App Storage	Long-term storage for user accounts, chats, document identifiers, document metadata, and document files.
OCR	The process of converting text from scanned or photographed documents into machine-readable text before downstream extraction.

Sprint #	Sprint goal
1	None
2	None
3	None
4	Refine the knowledge graph pipeline, set up the database for persistent app storage, and implement a consistent layout.
5	Implement user authentication and backend integration for document uploads while building the frontend UI components and backend pipeline required for knowledge graph extraction.
6	Complete the end-to-end application by integrating frontend views with backend APIs, enabling knowledge graph extraction and providing visualisation options for extracted knowledge graphs.
7	Improve knowledge graph quality, introduce AI chat capabilities, and enhance the user experience with document handling and usability improvements.
8	Create a unified user knowledge graph and connect it to the chat system, enabling context-aware conversations and persistent chat management.
9	
10	
11	
12	
13	
14	
15	

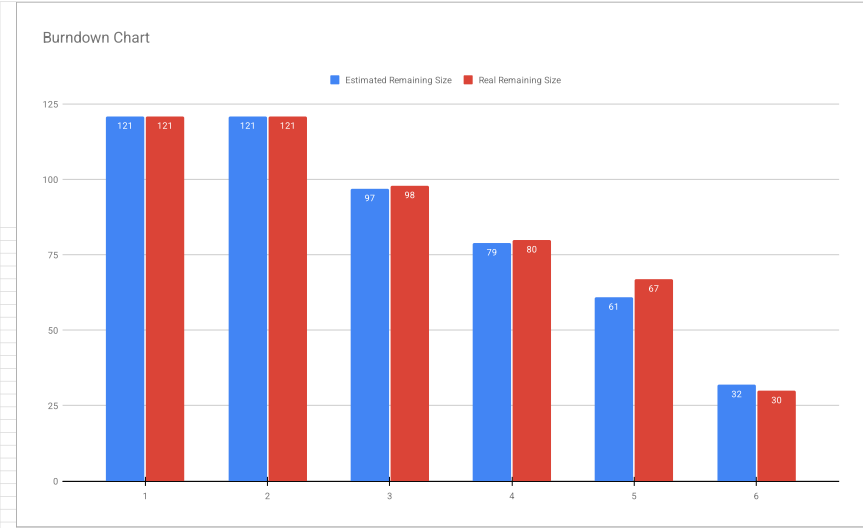
Sprint #	Story Points Realized	Average
1		23.4
2	23	23.4
3	18	23.4
4	13	23.4
5	37	23.4
6	26	23.4
7		23.4
8		23.4
9		23.4
10		23.4
11		23.4
12		23.4
13		23.4
	PLEASE CREATE THE VELOCITY CHART ON A NEW TAB USING THE DATA FROM THIS TAB	

Velocity Chart



Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			121	121		
Sprints						
1	Establish the project foundation by completing initial research, defining the app direction, and setting up the frontend development environment.		0	121	0	121
2	Deliver the first functional application prototype with document upload, authentication UI, architecture planning, and initial document analysis capabilities.		24	121	23	121
3	Build the core document management UI, define the app navigation structure, and implement document capture using the camera while establishing the backend cloud infrastructure.		18	97	18	98
4	Refine the knowledge graph pipeline, set up the database for persistent app storage, and implement a consistent layout.		18	79	13	80
5	Implement user authentication and backend integration for document uploads while building the frontend UI components and backend pipeline required for knowledge graph extraction.		29	61	37	67
6	Complete the end-to-end application by integrating frontend views with backend APIs, enabling knowledge graph extraction and providing visualisation options for extracted knowledge graphs.		32	32	26	30
Features						
1	Establish the project foundation by completing initial research, defining the app direction, and setting up the frontend development environment.	Setup documentation README.md				
		Research which extraction tool do we use? (Document AI, OCR, OCR with Google AI)				
		App Design Mockups / Wireframes / Moodboard				
		Frontend Setup Development Environment for App				
2	Deliver the first functional application prototype with document upload, authentication UI, architecture planning, and initial document analysis capabilities.	Implement Document Analysis and Structured Data Output (JSON)	8		8	
		Deliverable: Software Architecture Design	8		8	
		Frontend: Github Actions CI/CD Pipeline	3		3	
		Interface includes a document upload screen	3		2	
		App Login Screen	2		2	
3	Build the core document management UI, define the app navigation structure, and implement document capture using the camera while establishing the backend cloud infrastructure.	Frontend: Documents Overview	3		5	
		Capturing Documents using Camera/Scanning	5		3	
		App Navigation Tree	5		5	
		Setup GCP	5		5	
4	Refine the knowledge graph pipeline, set up the database for persistent app storage, and implement a consistent layout.	Knowledge Graph Pipeline Refinement	8		5	
		Set up Database for App Persistent Storage	5		5	
		Consistent Layout	5		3	
5	Implement user authentication and backend integration for document uploads while building the frontend UI components and backend pipeline required for knowledge graph extraction.	User Registration (Sign Up) View	1		1	
		User Registration (Functionality)	3		2	
		Login Authentication Process	5		5	
		Knowledge Graph Download Button	2		3	

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
		UI for Knowledge Extraction	2		2	
		Extracting Knowledge Graphs from Documents	8		13	
		Backend API: Documents	5		8	
		GCP Scheduling Extraction	3		3	
6	Complete the end-to-end application by integrating frontend views with backend APIs, enabling knowledge graph extraction and providing visualisation options for extracted knowledge graphs.					
		Connecting Document Views to Backend	5		3	
		Connect Knowledge Extraction UI to Backend	5		3	
		Connecting Document Upload to Backend	3		5	
		Home Screen with Appropriate Navigation Options	3		5	
		Simple Combining Knowledge Graphs	8		5	
		Backend Deployment: Knowledge Graph Components	8		5	
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				

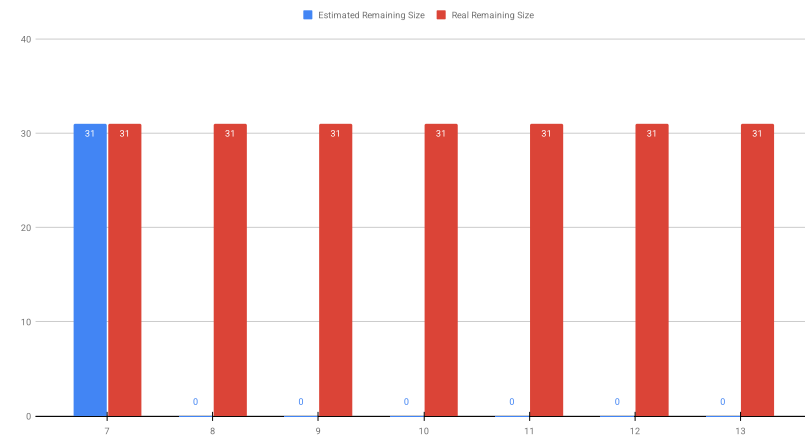


Fügen

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			31	31		
Sprints						
7	Improve knowledge graph quality, introduce AI chat capabilities, and enhance the user experience with document handling and usability improvements.		31	31	0	31
8	Create a unified user knowledge graph and connect it to the chat system, enabling context-aware conversations and persistent chat management.		0	0	0	31
9	Introduce voice call chat mode and finalise knowledge graph pipeline.		0	0	0	31
10	Integrate external knowledge sources such as research papers as AI chat contexts and visualise clickable sources in AI answers.		0	0	0	31
11	Implement normal RAG chat as comparison to knowledge graph contexts in AI chats.		0	0	0	31
12	Finalise features, documentation, bug fixes and polishing of UX.		0	0	0	31
13			0	0	0	31
Features						
7	Improve knowledge graph quality, introduce AI chat capabilities, and enhance the user experience with document handling and usability improvements.	Fix: Knowledge Graph Extraction API	3			
		Knowledge Graph Refinement	5			
		API: LLM Chat Agent	5			
		Chat Screen	5			
		Deleting Documents	3			
		Fix: Account Signup	3			
		Preview of Uploaded Documents	2			
		Componentisation - Technical Debt - Beautify	5			
8	Create a unified user knowledge graph and connect it to the chat system, enabling context-aware conversations and persistent chat management.	Ensure Knowledge Graph Correctness				
		Combining all document sub-graphs into one				
		API: Retrieving the Knowledge Graph				
		Knowledge Graph Context into LLM Prompts				
		API Test Environment				
		Creating Chats				
		Persistent Chat History				
		Connect Chat Screen with LLM Chat Agent				
		Chat Overview				
9	Introduce voice call chat mode and finalise knowledge graph pipeline.	Voice Chat Backend Endpoint				
		LLM Live API Connection				
		Voice Chat Indicator				
		Transcribing Microphone Input				
		Integrating Camera Scanning				
10	Integrate external knowledge sources such as research papers as AI chat contexts and visualise clickable sources in AI answers.	Scraping external research papers				
		Ingest external knowledge sources into LLM context / knowledge graph				
		Sources for an LLM Answer				
		Displaying LLM answer sources				
		Filtering Chats				
		Filtering Documents				
11	Implement normal RAG chat as comparison to knowledge graph contexts in AI chats.	Documents RAG LLM context				
		Switch between Knowledge Graph and RAG Context				
		Design Test Study for Approach Comparison				

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
		Written Report Comparing Approaches on Test Data				
12	Finalise features, documentation, bug fixes and polishing of UX.	Update User, Build and Design documentation				
		Code Cleanup and documentation				
13						
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				

Burndown Chart



Fügen

[illegible]

[illegible]

#	Context	Name	Version	License	Comment
	frontend	@react-navigation/bottom-tabs	^7.15.5	MIT	also counts for all expo subpackages
	frontend	@react-navigation/elements	^2.9.10	MIT	
	frontend	@react-navigation/native	^7.1.33	MIT	
	frontend	@tamagui/config	^2.0.0-rc.41	MIT	
	frontend	@tamagui/lucide-icons-2	^2.0.0-rc.41	MIT	
	frontend	expo	~55.0.15	MIT	
	frontend	expo-constants	~55.0.14	MIT	
	frontend	expo-device	~55.0.15	MIT	
	frontend	expo-document-picker	~55.0.13	MIT	
	frontend	expo-font	~55.0.6	MIT	
	frontend	expo-glass-effect	~55.0.10	MIT	
	frontend	expo-image	~55.0.8	MIT	
	frontend	expo-image-picker	~55.0.20	MIT	
	frontend	expo-linking	~55.0.13	MIT	
	frontend	expo-router	~55.0.12	MIT	
	frontend	expo-splash-screen	~55.0.18	MIT	
	frontend	expo-status-bar	~55.0.5	MIT	
	frontend	expo-symbols	~55.0.7	MIT	
	frontend	expo-system-ui	~55.0.15	MIT	
	frontend	expo-web-browser	~55.0.14	MIT	
	frontend	react	19.2.0	MIT	
	frontend	react-dom	19.2.0	MIT	
	frontend	react-hook-form	^7.73.1	MIT	
	frontend	react-native	0.83.4	MIT	
	frontend	react-native-gesture-handler	~2.30.0	MIT	
	frontend	react-native-reanimated	4.2.1	MIT	
	frontend	react-native-safe-area-context	~5.6.2	MIT	
	frontend	react-native-screens	~4.23.0	MIT	
	frontend	react-native-svg	15.15.3	MIT	
	frontend	react-native-web	~0.21.0	MIT	
	frontend	react-native-worklets	0.7.2	MIT	
	frontend	tamagui	^2.0.0-rc.41	MIT	
	frontend	@types/react	~19.2.2	MIT	
	frontend	eslint	^9.0.0	MIT	
	frontend	eslint-config-expo	~55.0.0	MIT	
	frontend	eslint-config-prettier	^10.1.8	MIT	
	frontend	eslint-config-universe	^15.0.3	MIT	
	frontend	eslint-plugin-prettier	^5.5.5	MIT	
	frontend	prettier	^3.8.3	MIT	
	frontend	typescript	~5.9.2	Apache-2.0	
	frontend	jotai	^5.100.9	MIT	

#	Context	Name	Version	License	Comment
	frontend	@tanstack/query	^2.20.0	MIT	
	frontend	expo-linear-gradient	~55.0.14	MIT	
	frontend	expo-clipboard	~55.0.13	MIT	
	backend	neo4j	6.1.0	Apache 2.0 AND Python-2.0	
	backend	numpy	2.2.6	BSD License	
	backend	openai	2.34.0	Apache Software License	
	backend	opencv-python	4.12.0.88	Apache Software License	
	backend	paddleocr	3.5.0	Apache License 2.0	
	backend	pandas	3.0.2	BSD License	
	backend	pillow	12.2.0	MIT-CMU	
	backend	pydantic	2.13.3	MIT	
	backend	python-dotenv	1.2.2	BSD-3-Clause	
	backend	requests	2.33.1	Apache Software License	
	backend	protobuf	3.20.2	BSD-3-Clause	
	backend	fast-api	0.136.1	MIT	
	backend	uvicorn	0.46.0	BSD-3-Clause	
	backend	firebase-admin	7.4.0	Apache-2.0	
	backend	google-cloud-firestore	2.27.0	Apache-2.0	
	backend	python-multipart	0.0.27	Apache-2.0	

Last Name	First Name	Value					
Brüggemann	Jonas	0		0.00	OK		
Khan	Muhamad	0					
Ebied	Andrew						
Vinzens	Hannes						
Ahmed	Hasnat			0	No size		
Amer	Hossam			1	Trivial size		
Sandt	Eloi			2	Small size		
Busse	Tobias			3	Medium size		
Zeeshan	Muhamad			5	Large size		
Haseeb	Abdul			8	Very large size		
				13	Too large (size)		
How to play planning poker							
1. Everyone type their number into their value field, don't hit return yet							
2. Someone, perhaps a product owner, count down 3.. 2.. 1..							
3. Then, everyone hit return to submit their value							